

Subliminal Persona Transfer in Language Models

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next-level-personas

1 Literature Review

1.1 Persona Vectors

A foundational insight in mechanistic interpretability is that high-level behavioral traits are encoded as approximately linear directions in activation space, which can be both read by projection and written by adding scaled copies to the residual stream (Chen et al., 2025). Chen et al. (2025) automate this into a pipeline that extracts a persona vector from any natural-language trait via contrastive prompting, taking the mean difference in residual activations between trait-promoting and trait-suppressing responses, and further show that finetuning on subtly malicious data, even unintentionally (e.g., insecure code), amplifies traits such as evil or sycophancy and degrades behavior in unrelated domains, mirroring the emergent misalignment of Betley et al. (2025).

Persona shift can also occur without any finetuning: Lu et al. (2026) embed hundreds of character archetypes via PCA and identify a dominant "Assistant Axis" along which models drift during multi-turn or emotionally charged interactions, with role-play and jailbreaks more frequent on its negative side. More broadly, representation engineering (Zou et al., 2023) casts such traits as structured, actionable directions: Rinsky et al. (2024) steer outputs by injecting activation vectors, and Li et al. (2023) improve truthfulness through targeted interventions without retraining. Together, these results establish that persona is geometrically structured, controllable, and causally linked to behavior, raising the question of whether it remains confined to a single model.

1.2 Subliminal Learning

Recent work suggests that behavioral traits may be transmitted even when they are not explicitly present in the data. Cloud et al. (2025) show that language models can transmit behavioral traits

through the statistical structure of generated data, even when the data is semantically unrelated to those traits. This phenomenon, known as subliminal learning, indicates that hidden signals in data distributions can encode and propagate biases.

Subsequent work studies how this transfer occurs. Schrodi et al. (2025) analyze when hidden biases transfer, while Morgulis and Hewitt (2026) show that these signals can be strengthened through specific encoding strategies. Complementary findings by Burns et al. (2022) suggest that important information may remain hidden within model activations even when it is not reflected in outputs, supporting the idea that traits can be transmitted implicitly.

1.3 Emergent Misalignment

The possibility of implicit transfer raises concerns for alignment. Betley et al. (2025) show that finetuning on narrow misaligned data, such as insecure code, can lead to broadly misaligned behavior across unrelated domains. Wang et al. (2025) give a mechanistic account, identifying a toxic persona feature in activation space that causally controls this effect. Wyse et al. (2025) further show that such misalignment is highly prompt-sensitive. Together, these results emphasize the fragility of model behavior and the risks of unintended shifts.

What remains unexplored, however, is whether a persona can propagate subliminally from teacher to student through semantically unrelated data, the question our project directly addresses. Prior work shows that behavioral traits are structured and controllable, can be transmitted implicitly through data, and that small shifts can lead to broad misalignment. Together, these results suggest that subliminal channels may enable the transfer of complex persona representations. The monitoring and preventative-steering tools of Chen et al. (2025) then provide a natural framework for detecting and potentially mitigating such effects.

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